

Introduction to machine learning - Spring 2025

Home Assignment 1

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1 Correlation and independence

- A. We want to prove that if X and Y are **independent**, then they are **uncorrelated**, meaning $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Case 1: X is discrete.

We start by looking at $\mathbb{E}[XY], \mathbb{E}[X], \mathbb{E}[Y]$

$$\mathbb{E}[XY] = \sum_x \sum_y xy \mathbb{P}(X = x, Y = y) \quad (1)$$

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x) \quad (2)$$

$$\mathbb{E}[Y] = \sum_y y \mathbb{P}(Y = y) \quad (3)$$

Since X and Y are independent, their joint probability factorizes.

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y). \quad (4)$$

from (4) and (1) we get:

$$\mathbb{E}[XY] = \sum_x \sum_y xy \mathbb{P}(X = x) \mathbb{P}(Y = y)$$

We arrange it so that each sum corresponds to its own variable

$$\mathbb{E}[XY] = \sum_x x \mathbb{P}(X = x) \sum_y y \mathbb{P}(Y = y) \quad (5)$$

from (2), (3) and (5) we get $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Case 2: X is continuous.

Now the expectations are with integrals

$$\mathbb{E}[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy \quad (6)$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (7)$$

$$\mathbb{E}[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy \quad (8)$$

Since X and Y are independent, their joint probability density function factorizes.

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y) \quad (9)$$

from (9) and (6) we get:

$$\mathbb{E}[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) \cdot f_Y(y) dx dy$$

We separate the integrals

$$\mathbb{E}[XY] = \int_{-\infty}^{\infty} x f_X(x) dx \int_{-\infty}^{\infty} y f_Y(y) dy \quad (10)$$

from (7), (8) and (10) we get $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ which completes the proof.

- B. For example, we can take $X \sim U(-1, 1)$ and $Y = X^2$ they are clearly not independent since knowing X fully determines Y but they are uncorrelated. We can verify that by calculating. $E[X] = 0$ and $E[Y] = \frac{1}{2} \int_{-1}^1 x^2 dx = \frac{1}{3}$, $E[XY] = E[X^3] = \frac{1}{2} \int_{-1}^1 x^3 dx = 0$ and we see that indeed they are uncorrelated because $E[X]E[Y] = E[XY]$.
- C. $X \sim \text{Bernoulli}(0.5), X \in \{0, 1\}$
 $Y \sim \text{Bernoulli}(0.5), Y \in \{0, 1\}$
 $Z = X \oplus Y$
 X and Y are independent by definition, and X, Z (and similarly Y, Z) are independent because knowing one does not give any information on the other as it could be any value with equal probability. And any variable can be completely determined if the two other variables are known because we can XOR any two values and know the 3rd.
- D. We will try to see how $z^T C z$ looks like

$$z^T C z \stackrel{\text{definition}}{=} z^T \left(\sum_{i=1}^n x_i x_i^T \right) z \stackrel{\text{rearrange}}{=} \sum_{i=1}^n z^T x_i x_i^T z \stackrel{\text{grouping}}{=} \sum_{i=1}^n (z^T x_i)(x_i^T z)$$

And because $z^T x_i$ is a scalar $z^T x_i = (z^T x_i)^T = x_i^T z$ we get

$$z^T C z = \sum_{i=1}^n (x_i^T z)^2$$

and since $(x_i^T z)^2 \geq 0$ for every i , the sum will also be non negative, which means C is positive semi-definite!

2 Multivariate Gaussians

- A. For $d = 1$ we get $x \sim \mathcal{N}(0, 1)$ and since the standard deviation is 1 we want a sample of $x \in [-1, 1]$. To calculate this probability, we will use cumulative distribution function (CDF).

$$P(-1 \leq X \leq 1) = CDF(1) - CDF(-1)$$

This normal distribution is symmetric so $CDF(-1) = 1 - CDF(1)$ and we get

$$P(-1 \leq X \leq 1) = 2CDF(1) - 1$$

From standard normal tables we know that $CDF(1) \simeq 0.8413$ and we get that

$$P(-1 \leq X \leq 1) \approx 0.6826$$

So the probability that a sample is within one standard deviation from the mean (zero) is 68.26%

- B. In class we saw that with the special case of standard multivariate Gaussian, the X_i 's are independent. We have $X \sim \mathcal{N}(\vec{0}, I_d)$ which is d distributions, each $X_i \sim \mathcal{N}(0, 1)$. We want to prove that the probability of the sample being within the unit sphere around zero (i.e $P(\|X\| \leq 1)$) is decaying to zero. Using the hint that the d -dimensional unit sphere is contained within a d -dimensional cube around zero with an edge side of 2 we will calculate the probability to be within the cube which is $P(\forall_{1 \leq i \leq d} X_i \in [-1, 1])$ and since the sphere is within the cube we know that $P(\|X\| \leq 1) \leq P(\forall_{1 \leq i \leq d} X_i \in [-1, 1])$ and since the X_i 's are independent the probability to be in the cube is $\prod_{i=1}^d P(X_i \in [-1, 1])$ which gives us 0.6826^d and as d goes to infinity $\lim_{d \rightarrow \infty} 0.6826^d = 0$ which means we got an upper bound that goes to zero so $P(\|X\| \leq 1) \rightarrow 0$.

3 Linear transformation of the covariance matrix

We start with variance definition

$$Var(y) = E[(y - E[y])^2] \stackrel{y=w^T x}{=} E[(w^T X - E[w^T X])(w^T X - E[w^T X])^T]$$

Factor out constants from inner E we get

$$Var(y) = E[(w^T X - w^T E[X])(w^T X - w^T E[X])^T]$$

Apply the transpose

$$Var(y) = E[(w^T X - w^T E[X])(X^T w - E[X]^T w)]$$

Separating the constants

$$Var(y) = E[w^T (X - E[X])(X^T - E[X]^T) w]$$

Taking the constants out of the E

$$Var(y) = w^T E[(X - E[X])(X^T - E[X]^T)] w$$

Rephrasing the 2nd term inside the E

$$Var(y) = w^T E[(X - E[X])(X - E[X])^T] w$$

From the lecture we know that $C = E[(X - E[X])(X - E[X])^T]$ and thus $Var(y) = w^T C w$

4 Eigen decomposition

- a. It is enough to show that the characteristic polynomials of A and A^T are the same to determine they have the same eigen values.

We know that for A the characteristic polynomial is $\det(A - \lambda I)$ and for A^T it is $\det(A^T - \lambda I)$.

We know that $\det(B) = \det(B^T)$ for every matrix B . so

$$\det(A - \lambda I) = \det((A - \lambda I)^T) \quad (11)$$

Knowing that a transpose of a scalar is the scalar and the transpose of the identity matrix is the identity matrix, If we apply the transpose we see that

$$(A - \lambda I)^T = A^T - \lambda I \quad (12)$$

From (11) and (12) we get that

$$\det(A - \lambda I) = \det(A^T - \lambda I)$$

Which means their eigenvalues are the same!

- b. if A is diagonalizable we know it can be expressed in the form

$$A = P D P^{-1}$$

The eigenvectors form the columns of P , and the corresponding eigenvalues are the entries on the diagonal of D .

$$\det(A) = \det(P D P^{-1}) = \det(P) \det(D) \det(P^{-1})$$

and since we know that $\det(P^{-1}) = \frac{1}{\det(P)}$ so we get that

$$\det(A) = \det(P) \det(D) \frac{1}{\det(P)} = \det(D)$$

and since D is diagonal the determinant is just the multiplication of the values so

$$\det(A) = \det(D) = \prod_{i=1}^n \lambda_i$$

5 Gradients

For a function to be differentiable it must be smooth and continuous.

- a. First we see that for $w = \vec{0}$ we get $f(w, x) = 1$ which is differentiable for every x and $\nabla_x f = \vec{0}$.
 Let's focus on the more interesting case where $w \neq \vec{0}$, we see a split function that consists of two elementary functions, the zero function and a linear function $1 - w^T x$, at points which are not on the boundary where $w^T x = 1$ it is trivial and differentiable. and if we focus on $w^T x = 1$ we see that it goes to 0 from both sides which makes it continuous.
 Looking at the derivative of $w^T x$ w.r.t x

$$\frac{\partial w^T x}{\partial x_i} = \frac{\partial \sum_{j=1}^d w_j x_j}{\partial x_i} = w_i$$

The derivative would be

$$\nabla_x f = \begin{cases} 0 & w^T x > 1 \\ -w & w^T x < 1 \end{cases}$$

Since $w \neq \vec{0}$ we see that $\nabla_x f$ when $w^T x = 1$ is not the same for both sides which means it's not differentiable at this point. so in conclusion we can say its differentiable at $\{x \in \mathbb{R}^d \mid w^T x \neq 1\}$

- b. It's a composition of elementary functions and the denominator is always positive. its differentiable at any $x \in \mathbb{R}$.

We remember that $\frac{\partial w^T x}{\partial x_i} = w_i$ so the derivative would be

$$\frac{\partial f}{\partial x} = ((1 + e^{-w^T x})^{-1})' = -(1 + e^{-w^T x})^{-2} e^{-w^T x} (-w) = \frac{w e^{-w^T x}}{(1 + e^{-w^T x})^2}$$

Let's show that $\nabla_x f(x) = f(x)(1 - f(x))w$

$$1 - f(x) = 1 - \frac{1}{1 + e^{-w^T x}} = \frac{e^{-w^T x}}{1 + e^{-w^T x}}$$

$$f(x)(1 - f(x))w = \left(\frac{1}{1 + e^{-w^T x}}\right) \left(\frac{e^{-w^T x}}{1 + e^{-w^T x}}\right)(w) = \frac{w e^{-w^T x}}{(1 + e^{-w^T x})^2}$$